



Solving the Lorenz ODE System using **Optimal ANN** Architectures

Can a **plain feedforward ANN**, with no physics in the loss, learn a coupled nonlinear ODE system, and **which architecture does it best?**

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01 The Problem



Classical solver

Step-by-step, mesh-bound, re-runs per query

vs



Trained ANN

One direct map $t \rightarrow (x, y, z)$, mesh-free

Central question: can a plain ANN, trained only on data with no physics in the loss, learn this coupled system — and **which architecture does it best?**

02 Objectives



Trusted baseline

RK4 + SciPy, cross-checked



Search the space

Depth $\times$ width $\times$ activation



Accuracy vs cost

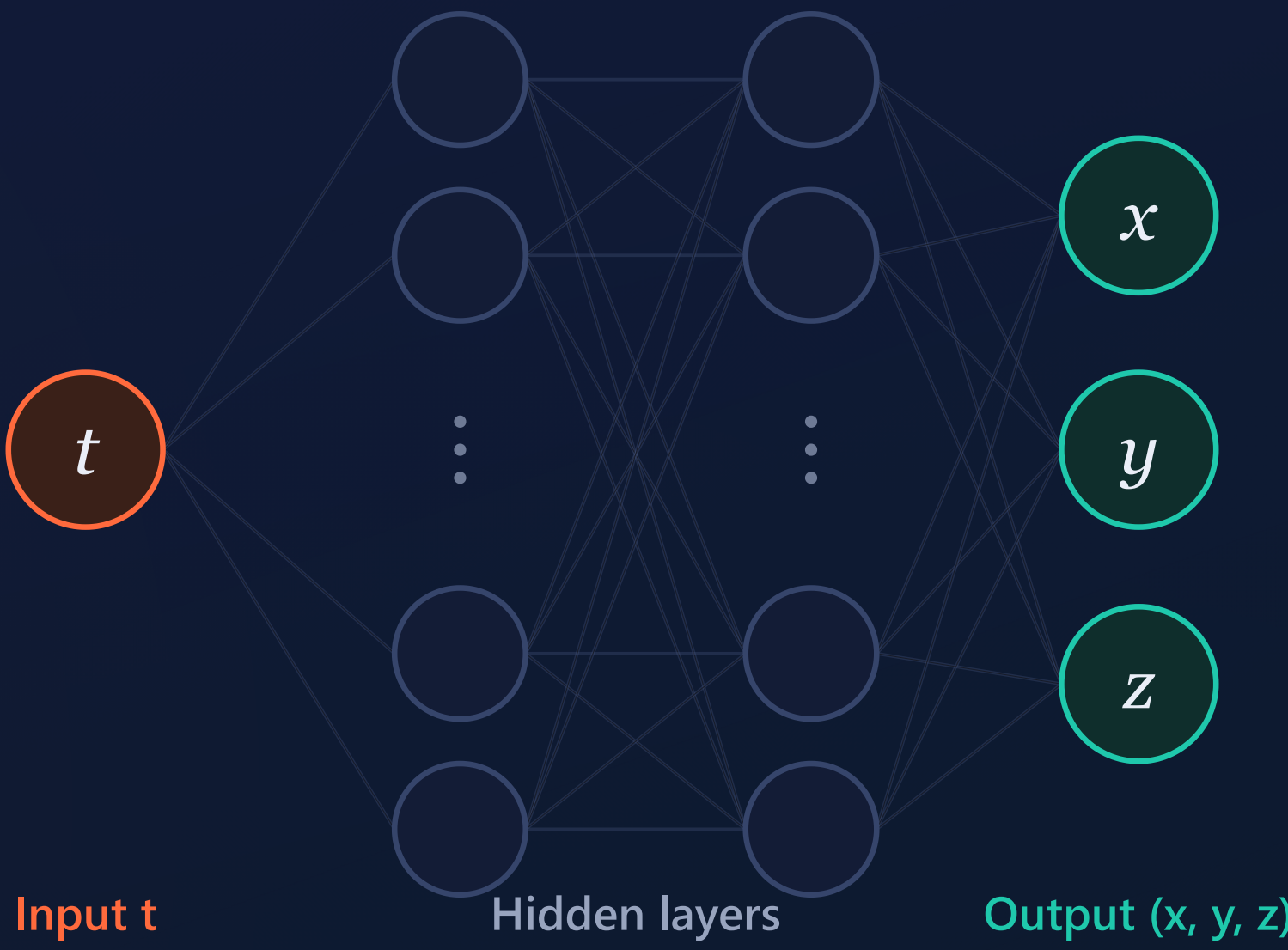
Error and run-time, together



Benchmark

Best ANN vs PINN literature

04 Network & Search Space



Depth

LAYERS

1

2

3

4

$\times$

Width

NEURONS

20

50

100

$\times$

Activation

NONLINEARITY

tanh

ReLU

sigmoid

GELU

Swish

= **60** candidate architectures

03 The System

REDUCED FORM · $K = 2, L = 1$

$$\dot{x} = -0.10 y z$$

$$\dot{y} = +1.60 x z$$

$$\dot{z} = -0.75 x y$$

INITIAL

x_0, y_0, z_0

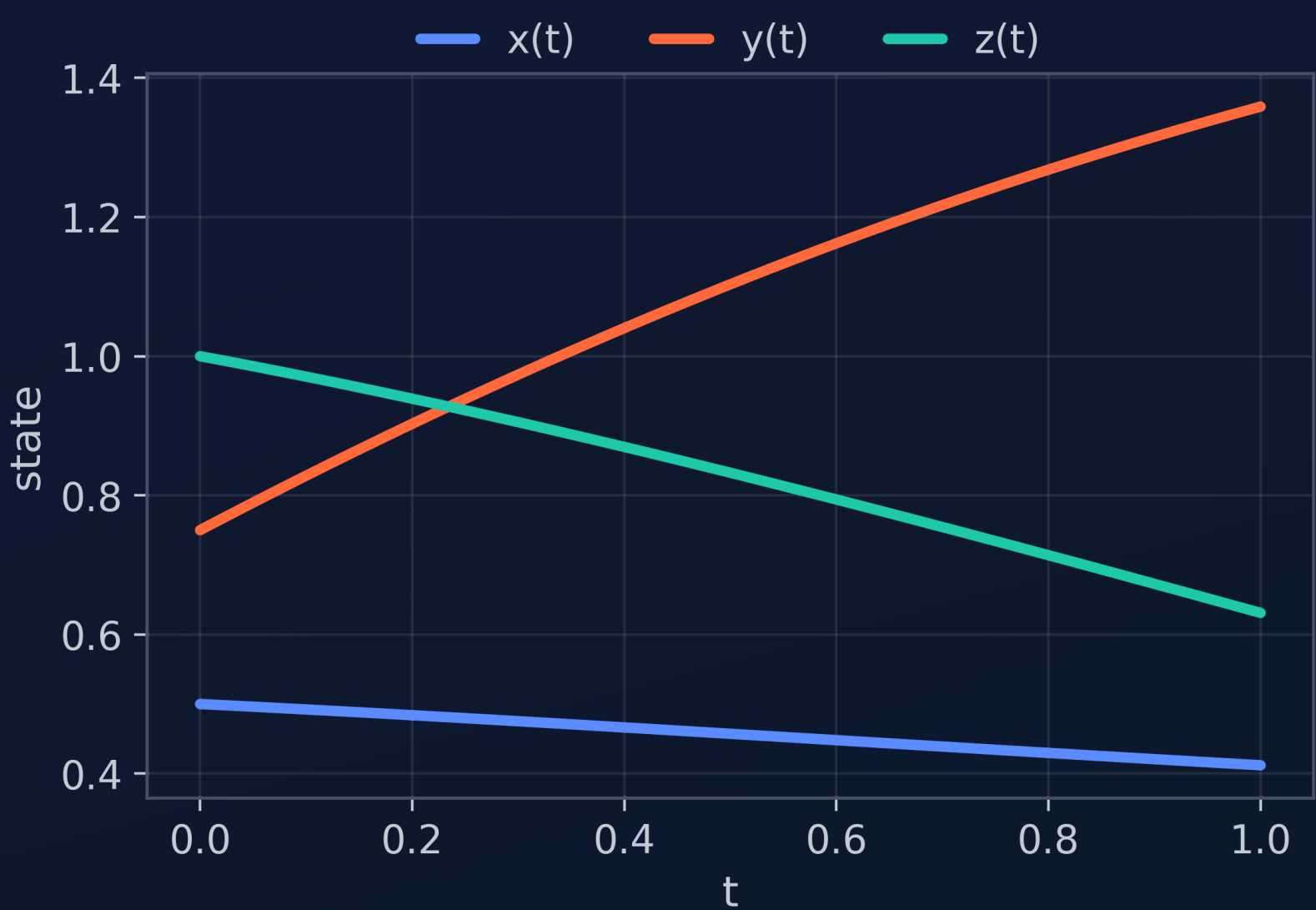
VALUES

0.5, 0.75, 1

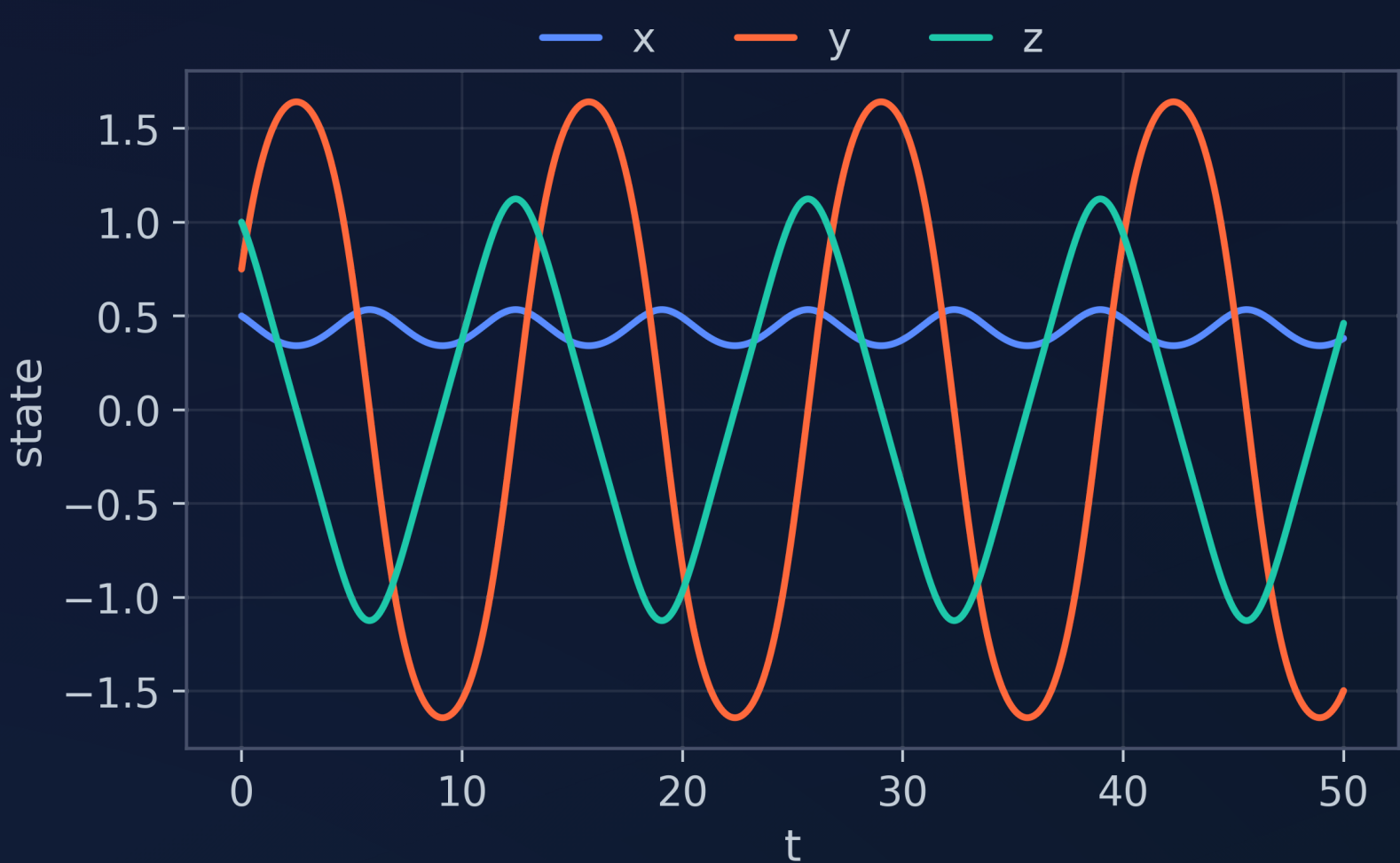
INTERVAL

$t \in [0, 1]$

Three coupled nonlinear ODEs: simple to state, hard to integrate accurately.



RK4 reference solution on $t \in [0, 1]$: the data the ANN learns.



Long-term behaviour on $t \in [0, 50]$: bounded, quasi-periodic orbits.

05 Two-Phase Study

PHASE 1

60

architecture search · Adam fixed

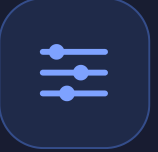
PHASE 2

9

optimizer sweep · top-3 nets

Architecture first, optimizer second; the two effects stay separable.

→ Pipeline



Generate

RK4 + SciPy reference

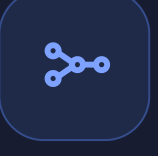
↓



Split

80 / 20 train-validation

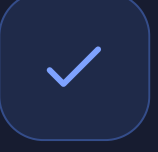
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Train

69 controlled runs

↓



Select

best architecture

06 How We Score



MSE

mean sq. error



MAE

mean abs. error



Max error

worst case



Train time

fitting cost



Inference time

cost of a single query once trained

07 The Gap

PRIOR WORK

LORENZ-1960

PLAIN ANN

ARCH. SWEEP

Matthews & Bihlo

✓

✗

✗

Lagaris et al.

✗

✓

✗

Lau & Werth

✗

✓

✗

Mall & Chakraverty

✗

✓

✗

Ours · Team_ParaDox

✓

✓

✓

No prior study sweeps plain-ANN architecture on Lorenz-1960.

08 Baseline Verified

target $< 10^{-10}$

1.8×10^{-22}

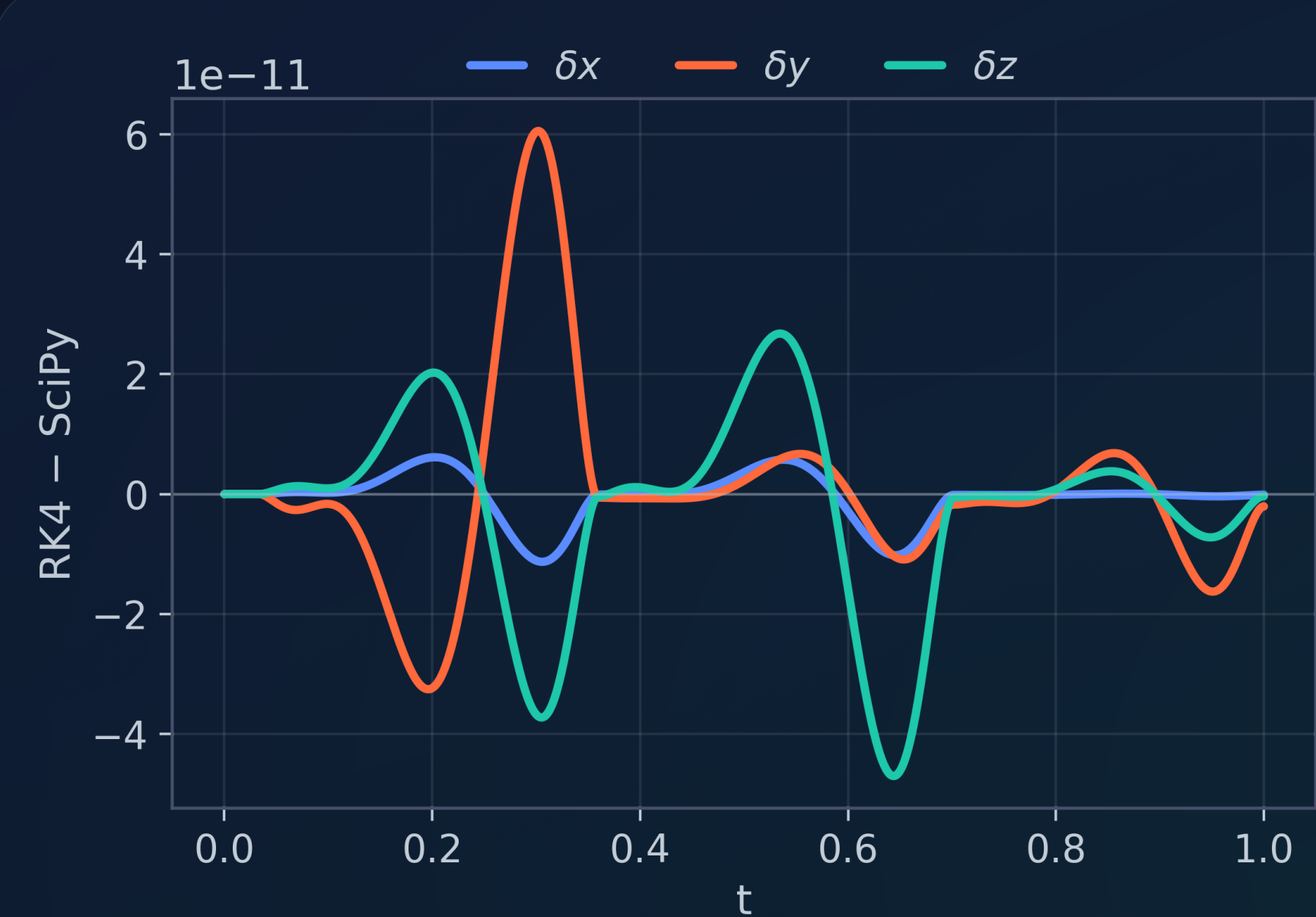
solver-agreement MSE (RK4 vs SciPy)

6.06×10^{-11}

max pointwise disagreement

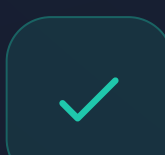
✓ target beaten by ten orders of magnitude

Two independent solvers agree before any ANN is trained.



Pointwise RK4 - SciPy disagreement across the trajectory.

09 Status & Next



FYDP-I · done

17-paper review, verified dual-solver baseline, locked 69-run design



FYDP-II · next

Run the experiments, repeat seeds, benchmark against PINNs

No claims yet. Accuracy results follow once FYDP-II runs.